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	POSTDOCTORAL FELLOWSHIP. Space radiation presents a significant hazard to space flight crews. It is dominated by high energy charged particles such as protons and heavy ions, which may cause significant DNA damage at relevant exposure levels. Depending on the radiation dose, this DNA damage can range from an extent that is compatible with cell survival to levels that induce cell death. The latter can cause a potentially life-threatening decrease in radiation-sensitive cells such as white blood cells and platelets (anemia), the former may manifest itself in malignant transformation of irradiated cells, which can ultimately lead to cancer. In the past few years, a group of enzymes called the Sir2-class of histone deacetylases (class III HDACs) or "the sirtuins" have been shown to regulate cell defenses against DNA damage and biological stress. Sirtuins may further underlie the stress resistance and increased lifespan of mammals fed a calorie-restricted diet. In simple organisms such as yeast, worms and flies, additional copies of the Sir2 gene make cells more resistant to		

Task Description:	radiation and extend the organism's lifespan by ~30%. In rodents, calorie-restriction causes increased radiation resistance and a significantly lower cancer incidence, as well as lifespan extension. Yeast and fly data show that sirtuins are required for the maintenance of genomic integrity, in particular that of highly repetitive, tightly packed “heterochromatin”. Loss of this type of chromatin can cause dramatic genomic changes, affecting cell function and possibly survival. Yeast sirtuins were further shown to directly aid the repair of DNA damage. While the precise role played by sirtuins in higher organisms remains elusive, published results demonstrate that upregulation of sirtuins leads to numerous cytoprotective events including increased cell survival and activation of DNA repair pathways. The goal of this proposal is to elucidate the mechanisms by which mammalian sirtuins may confer genomic integrity in response to genotoxic stress both in cell culture and in animals. Specifically, we were interested in investigating (1) the ability of sirtuins to regulate heterochromatin and thereby genomic stability, (2) whether sirtuins can protect from chromosomal instability in response to DNA damage, and (3) the impact of increased sirtuin activity (through overexpression or pharmacological activation) on cancer development in mice. Data obtained in this study suggest a conserved role for sirtuins in the maintenance of genomic integrity and protection from DNA damage, in organisms as diverse as yeast and mammals. Specifically we found that, in yeast, oxidative stress results in a loss of silencing and increased DNA instability at loci that are normally repressed by Sir2. Paralleling yeast Sir2, its mouse orthologue SIRT1 was shown to participate in the silencing of repetitive (pericentromeric) DNA. This silencing was reduced upon exposure to radical oxygen species (ROS). By mapping the promoters that SIRT1 binds to, we show that this loss of silencing is not limited to repetitive DNA but extends to other SIRT1-associated genomic loci. This major change in chromatin-associated SIRT1 upon exposure to DNA damage appears to be driven by recruitment of SIRT1 to sites of DNA damage. Consistently, SIRT1 is required for efficient DNA repair and lack of SIRT1 increases genomic instability in response to oxidative stress. This redistribution of SIRT1 in response to DNA damage comes, however, at the cost of epigenetic changes at SIRT1-regulated genomic loci, involving deregulation of gene expression and loss of repeat silencing. A number of SIRT1-mediated epigenetic changes observed in vitro are recapitulated in the aging brain and can, furthermore, contribute to neuronal decline, suggesting a dichotomy of SIRT1 function that may contribute to normal aging in mammals. Importantly, increased SIRT1 activity was shown to reduce ROS-induced cellular changes and may therefore be a way to decrease the radiation risk with regard to both genomic stability and potentially detrimental changes in gene expression. To mimic increased sirtuin activity in animals, a mouse model was generated that allows for the inducible overexpression of SIRT1. Using an irradiation-induced mouse cancer model, we found that increased expression of SIRT1 extends survival and delays the onset of radiation-induced cancer, in particular that of thymic lymphomas. Importantly, sirtuin activity can be enhanced using sirtuin-activating compounds (STACs) that we have discovered, which makes sirtuins an ideal target for pharmacological protection from radiation-induced pathologies during or following space flight. We show that the STAC resveratrol has a similar protective effect on tumorigenesis as observed in mice expressing higher levels of SIRT1. Together, our data demonstrate a conserved, protective role for SIRT1 during genotoxic stress. In conjunction with the availability of a number of STACs, SIRT1, therefore, presents a promising candidate for pharmacological intervention during (and after) exposure to potentially hazardous space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In the past few years, a group of enzymes called "sirtuins" have been shown to regulate cell defenses against damage and biological stress in diverse organisms, from yeast to mammals. They are thought to have evolved to help organisms survive adversity and may underlie the stress resistance and increased lifespan of mammals fed a calorie-restricted diet. In rodents, caloric restriction is one of the most effective ways to protect animals from radiation-induced DNA damage and cancer. Indeed, our lab has shown that calorie-restricted rats express elevated levels of the mouse sirtuin SIRT1 and that having additional copies of SIRT1 protects mammalian cells from radiation. In simple organisms such as yeast, worms and flies, additional copies of the sirtuin gene SIR2 gene make cells more resistant to radiation and extend average lifespan by ~30%. Thus, harnessing sirtuins may provide a means to greatly increase the survival of critical cells such as stem cells, white blood cells and neurons in response to radiation damage and possibly prevent radiation-induced cancers. Sirtuins can be activated using sirtuin-activating components (STACs) such as resveratrol, a small polyphenolic molecule produced by stressed plants. We further demonstrated that resveratrol promotes the survival of fibroblasts in response to irradiation. Our current research corroborates these findings and places SIRT1 in the context of DNA damage repair upon exposure to free radicals, a common and deleterious by-product of ionizing radiation. Radiation exposure is not limited to astronauts and our research may, therefore, provide benefits for professionals facing elevated radiation exposure levels such as airline personnel and radiation workers. Our research may further have important medical benefits for a large number of the of the US population. Free radicals are generated as part of daily living, as a consequence of respiration and energy metabolism. Free radical-induced damage is a major contributor to age-related organ decline and cancer. We explore the consequences of increased SIRT1 activity on free radical-induced damage. Specifically, it has been demonstrated recently that oxidative stress alters gene expression in the aging human brain. Deregulation of gene families involved in synaptic plasticity and other key neurobiological pathways were found to increase with age and correlate with oxidative stress. Our results extend these findings and show that oxidative stress mirrors a number of epigenetic changes normally observed in aged individuals. Some of these changes appear to be due to SIRT1 as increased SIRT1 activity can oppose these changes. This may be exploited to slow down the aging process by mitigating the consequences of oxidative stress on the human body. We envisage that STACs may find broad use in the US as a pharmacological agent against cancer, the age-related decline in brain function, and possibly aging itself.

Task Progress:	The goal of this study is to investigate the potential role of class III histone deacetylases, also known as 'sirtuins', in the maintenance of genomic stability and in the cellular defense against radiation damage. Given the deleterious effect of space radiation on genomic DNA, this proposal will provide useful insights into molecular mechanisms that may be utilized to protect space flight crews from the potentially detrimental consequences of exposure to space radiation. One of the deleterious consequences of space irradiation is the generation of reactive oxygen species (ROS) and their damaging impact on both cellular proteins and DNA. The latter can cause genomic instability and, ultimately, cancer. Our lab works on SIRT1-activating compounds (STACs) that might one day be taken by astronauts to protect against radiation damage. Data obtained in this study suggest a conserved role for sirtuins in the maintenance of genomic integrity and protection from DNA damage. W previously found a redistribution of chromatin-associated SIRT1 in response to genotoxic stress that resulted in a loss of silencing and at loci that are normally repressed by the yeast Sir2 or mammalian SIRT1. We now were able to show that SIRT1 is physically recruited to sites of DNA damage and is required for efficient DNA repair. Consistently, SIRT1-deficient embryonic stem cells are more susceptible to ROS-induced genomic instability. We furthermore show that the DNA damage-driven redistribution of SIRT1 can cause the deregulation of gene expression and loss of repeat silencing. A number of these epigenetic changes are recapitulated in the aging brain and can, furthermore, contribute to neuronal decline. Importantly, increased SIRT1 activity is able to counteract these changes. Together our data indicate that the genomic localization of SIRT1 changes during oxidative stress and thereby affects genomic integrity upon ROS exposure. To address whether these findings have an implication for radiation-induced pathologies such as cancer and anemia, we generated a mouse model that allows for temporally or spatially regulated SIRT1 overexpression. These mice were then crossed to a cancer model of genomic instability that is highly susceptible to radiation-induced DNA damage (mice lacking one copy of the tumor suppressor p53, termed p53+/- mice). Overexpression of SIRT1 resulted in increased survival after irradiation when compared to appropriate controls. Furthermore, the frequency of fatal thymic tumors was reduced twofold in SIRT1 transgenic animals resembling that of unirradiated p53+/- mice. This result provides a possible means to interfere with the development of (space) radiation-induced cancers. Similar to increased SIRT1 expression, the STAC resveratrol also delays tumor development in irradiated p53+/- mice, suggesting that resveratrol and other STACs may be used as a dietary countermeasure to mitigate cancer risks in response to radiation.
Bibliography Type:	Description: (Last Updated: 06/09/2008)
Articles in Peer-reviewed Journals	Oberdoerffer P, Sinclair DA. "The role of nuclear architecture in genomic instability and ageing." Nat Rev Mol Cell Biol. 2007 Sep;8(9):692-702. PMID: 17700626 , Sep-2007